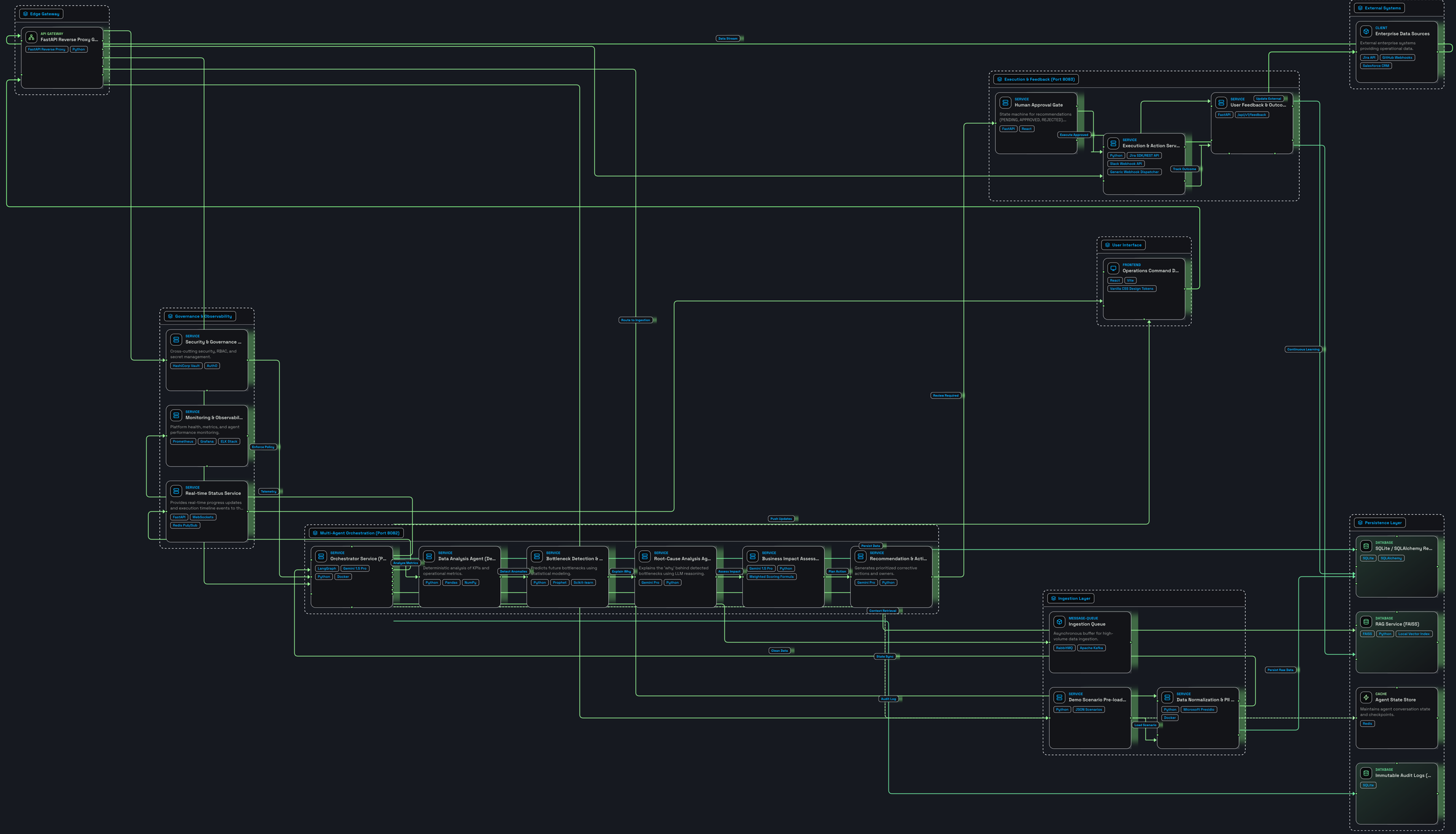
Edge Gateway
API GATEWAY
FastAPI Reverse Proxy G...
FastAPI Reverse Proxy
Python
Data Stream
External Systems
CLIENT
Enterprise Data Sources
External enterprise systems providing operational data.
Jira API
Github Webhooks
Salesforce CRM
Execution & Feedback (Port 8083)
SERVICE
Human Approval Gate
State machine for recommendations (PENDING, APPROVED, REJECTED)...
FastAPI
React
Execute Approved
SERVICE
Execution & Action Serv...
Python
Jira SDK/REST API
Slack Webhook API
Generic Webhook Dispatcher
Track Outcome
SERVICE
User Feedback & Outco...
Update External
FastAPI
/api/v1/feedback
User Interface
FRONTEND
Operations Command D...
React
Vite
Vanilla CSS Design Tokens
Governance & Observability
SERVICE
Security & Governance ...
Cross-cutting security, RBAC, and secret management.
HashiCorp Vault
Auth0
SERVICE
Monitoring & Observabil...
Platform health, metrics, and agent performance monitoring.
Prometheus
Grafana
ELK Stack
Enforce Policy
SERVICE
Real-time Status Service
Provides real-time progress updates and execution timeline events to th...
FastAPI
WebSockets
Redis Pub/Sub
Telemetry
Route to Ingestion
Review Required
Continuous Learning
Push Updates
Multi-Agent Orchestration (Port 8082)
SERVICE
Orchestrator Service (P...
LangGraph
Gemini 1.5 Pro
Python
Docker
Analyze Metrics
SERVICE
Data Analysis Agent (De...
Deterministic analysis of KPIs and operational metrics.
Python
Pandas
NumPy
Detect Anomalies
SERVICE
Bottleneck Detection & ...
Predicts future bottlenecks using statistical modeling.
Python
Prophet
Scikit-learn
Explain Why
SERVICE
Root-Cause Analysis Ag...
Explains the 'why' behind detected bottlenecks using LLM reasoning.
Gemini Pro
Python
Assess Impact
SERVICE
Business Impact Assess...
Gemini 1.5 Pro
Python
Weighted Scoring Formula
Plan Action
SERVICE
Recommendation & Acti...
Generates prioritized corrective actions and owners.
Gemini Pro
Python
Persist Data
Context Retrieval
Clean Data
State Sync
Audit Log
Persistence Layer
DATABASE
SQLite / SQLAlchemy Re...
SQLite
SQLAlchemy
DATABASE
RAG Service (FAISS)
FAISS
Python
Local Vector Index
CACHE
Agent State Store
Maintains agent conversation state and checkpoints.
Redis
DATABASE
Immutable Audit Logs (...
SQLite
Ingestion Layer
MESSAGE-QUEUE
Ingestion Queue
Asynchronous buffer for high-volume data ingestion.
RabbitMQ
Apache Kafka
SERVICE
Demo Scenario Pre-load...
Python
JSON Scenarios
Load Scenario
SERVICE
Data Normalization & PII ...
Python
Microsoft Presidio
Docker
Persist Raw Data